Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Prof. Dr. Jens Koenig	Email	jens.koenig@hs-esslingen.de
Institute Name	University of Applied Science Esslingen, Faculty Mobility and Technology			
Address	Kanalstraße 33, 73728 Esslingen, Germany			

2. Technology Overview

Title of Proposed Technology	*Novel Lightweight Solutions for Load Bearing Structures of Rail Vehicles*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicle	*Vehicle Body and light-weighting Technologies*		
Concept of Proposed Technology	*Railway and thus rail vehicles play a crucial role in the future mobility all over the world, especially in the highly developed societies where high-performance technology is requested (1). This requires continuous improvement and optimization. The extraordinary loads on railway vehicles (e.g. (8),(9)) need a comprehensive lightweight design, especially for highly loaded and stressed structures, such as the entire car body and its load bearing components (e.g. area between the coupler interface and the main cross beam). However, semi-structural parts, such as cladding elements or highly loaded beam elements, could a focus here. The concept of the proposed technology is the adaption and use of a novel and requirement adapted, comprehensive methodology for the lightweight optimization of a (semi-) load bearing component or section of a rail vehicle (10), (11), (12). The target is the significant weight reduction and the demonstration of the effect as a mock-up or demonstrator with the consideration of the manufacturing aspects and specific frame conditions.*			

(max 10 pages)

Rationale for Joint R&D Project	*Only a few countries worldwide play a leading role in railway vehicle development, among them South Korea and Germany, as well as Japan, France, Spain and, in recent years, China. Further progress beyond the already highly developed solutions can be achieved through the exchange and combination of technologies, methodologies and specific expertise for mutual benefit (exercised e.g. (1), (5), (6)).* *Here an established methodical approach for lightweight design of rail vehicles shall be taken, adapted and expended considering the specific frame conditions. The approach shall be combined with specific manufacturing aspects and used for the optimization of a (semi-)load bearing structure of a rail vehicle.* *The target is here to combining the methodical approach with specific manufacturing aspects, the demonstration of the performance of the comprehensive methodology for a weight reduction under changing conditions and increasing the knowledge. A significant weight reduction for a selected component or part of a railway vehicle (passenger or freight) could be demonstrated by manufacturing a demonstrator or prototype.*
Key Words	1. *Structurally and lightweight optimized structures of the car body and its components of rail vehicles* 2. *Comprehensive lightweight approach for highly stressed structures* 3. *Combination of lightweight design and manufacturing process for rail vehicles* 4. *Approach for the definition of the suitable material in the suitable area for semi-structural parts*

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (4-5)

R&D Objectives	*The objective of this joint R&D project is the lightweight optimization of a selected structural or semi-structural component or sub-structure of a railway vehicle car body or a part of it, using an advanced and requirement-adapted methodical lightweight design and optimization approach, considering manufacturing frame conditions. The results and effects of the methodology has been demonstrated a several times and in different topics; also in an industrial environment, national and international (e.g. (2), (3), (4), (6), (7).*
	The focus of the project is the systematic application, adaptation and use of a method on a component or structural section of a railway vehicle car body, such as a highly loaded interface region, beam structure, or cladding-related sub-structure. The result of the method is a lightweight optimized design a of structural or semi-structural part of a rail vehicle, considering the frame conditions like requirements, available manufacturing process, usable materials (only steel or aluminum or multi-material design, etc.). Based on the results, the selected structure is detailed, finalized, and manufactured. Afterwards a testing will be realized whereby the performance of the structure is validated. Thus, a systematically development chain shall be realized here and an approach which is transferable on other applications.
	The project is carried out jointly by a university partner and a South Korean SME, combining:
	• *The scientific and methodological expertise in structural design and lightweight optimization,*
	• *The industrial, application-oriented perspective and practical constraints of railway vehicle component development.*
	Within the project, the selected car body component or sub-structure will be:
	• *Analyzed under relevant railway-specific load cases and boundary conditions,*
	• *Optimized using a comprehensive lightweight design methodology*
	  ○ *Definition of the necessary frame conditions (requirement light weight design)*
	  ○ *Definition of the available space for the section or part and based on this topology optimizations for analyzing the load paths*
	  ○ *Systematical derivation of concepts of the structure considering the relevant frame conditions (conceptual light weight design, material lightweight design, lightweight design through functional integration)*
	  ○ *Method based selection of the most promising concept and detailing in CAD*
	  ○ *Pre-dimensioning and shape dimensioning using FE-Tools (shape and sizing lightweight design)*
	  ○ *Final design and dimensioning of the structure*
	The results will be validated by the design and engineering of a mock-up or demonstrator, showing the practical applicability and performance potential of the applied lightweight design methodology on a real railway vehicle structure or a part of it.			

(max 10 pages)

R&D Target Specifications	*The R&D project targets the lightweight optimization of a selected structural or semi-structural component or sub-structure of a rail vehicle. The target specifications are defined as follows:* • *Demonstration of significant weight reduction potential compared to a suitable reference, while considering structural, functional and regulatory requirements (tbd., e.g. 20-30%).* • *Application and use of a comprehensive lightweight design and optimization methodology on a representative rail car body component.* • *Verification of the structural performance of the optimized design using appropriate numerical simulation methods under relevant railway-specific load cases.* • *Design and engineering of a mock-up or demonstrator to illustrate the applicability and integration capability of the optimized structural concept.* *In addition, the project aims to:* • *Demonstrate the effectiveness of methodical lightweight optimization for railway car body structures,* • *Strengthen the transfer of knowledge between academia and industry*
R&D Plan	*Phase 1: Definition of Reference Case and Requirements: (~ 2 month)* • *Selection of a representative structural or semi-structural component or sub-structure of a rail vehicle car body* • *Analysis and Definition of frame conditions and requirement under consideration of the lightweight aspect (requirement lightweight design)* *Milestone 1: Definition of representative structure and frame conditions* *Phase 2: Structural Analysis and Baseline Assessment (~ 3 month)* • *Creation of numerical model* • *Structural assessment of the selected structure of the baseline under relevant load cases to evaluate weight, stiffness and strength characteristics* • *Definition of the frame conditions of the reference and optimization potential* *Milestone 2: frame conditions of the reference and optimization potential* *Phase 3: Lightweight Optimization and Concept Development (~ 12 month)* • *Redesign and adaption of the methodology to the specific requirements* • *Application of the methodical lightweight design and optimization approach using topology, shape and/or sizing optimization methods* • *Development of lightweight-optimized structural concepts based on the optimization results* • *Common evaluation of the conceptions of mechanical designs and selection of the most suitable* • *Pre-dimensioning and iterative evaluation of structural performance and lightweight potential* *Milestones 3: Definition of the most promising concept and pre-dimensioned concept*

(max 10 pages)

	Phase 4: Design Consolidation and Verification (~ 6 month) • *Detailing, design and engineering of the pre-dimensioned concept* • *Simulation and numerical verification of the lightweight structure* • *Data Preparation* *Milestone 4: CAD model and verification of the lightweight structure* *Phase 5: Demonstrator / Mock-up Design (~ 4 month)* • *Design and engineering of a mock-up or demonstrator of the optimized structure* • *Preparation of data for manufacturing* • *Manufacturing of lightweight structure* *Milestone 5: Physical demonstrator of the lightweight structure* *Phase 6: Evaluation and testing (~ 3 month)* • *Definition of the testing program* • *Testing of the demonstrator on testing facility* • *Comparison with the reference solution* *Milestone 6: Finished Testing with documentation* *Phase 7: Documentation and Knowledge Transfer (~ 2 month)* • *Common evaluation by the university and SME partners.* • *Documentation, main results and transferable findings* *Milestone 7: Final report* *Estimate duration 32 month (total)*

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Expected roles and responsibilities:* *Adaption and modification of the methodology, structural analysis, optimization methods, conceptions of the mechanical designs and common evaluation, pre-dimensioning of the structure, scientific evaluation.* *Research scope is the further development of the methodical approach and usage for the identified structure for the demonstration of the performance for weight reduction under specific (regional) frame conditions and requirements.*
Expected R&R of Korean SME Partner	*Expected roles and responsibilities:* *Provide application case, requirements, industrial perspective, detailing of the mechanical design based on the common selected concept ready for manufacture, manufacturing of the demonstrator and testing/validation of the demonstrator.* *Common coordination and work on different topics, discussions and exchange are crucial for the success of this project.* *Thus, regular exchange is necessary to ensure the purposeful proceeding. Effort depends on the project phase. In the average at least every 2 till 4 weeks as an online meeting. At least one till two times a face-to-face meeting in South Korea and Germany.* *One contact person is necessary which is able to make all relevant decision and coordinates the work of the engineer and the worker.*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The joint R&D project is expected to deliver a lightweight-optimized structural or semi-structural component of a railway vehicle car body and a validated engineering approach for its systematic development.* *Technological outcomes include:* • *Demonstration of a significant lightweight potential on a representative railway vehicle structure,* • *A simulation-validated structural concept meeting all relevant functional and structural requirements,* • *And a reproducible engineering workflow for methodical lightweight optimization of railway car body components.* *Commercialization potential arises from:* • *The direct applicability of the developed and validated lightweight concept to future vehicle or component developments of the industrial partner,* • *Improved product performance.* *Broader industrial and economic impact includes:* • *Improved energy efficiency and sustainability of railway vehicles through weight reduction,* • *Strengthened competitiveness of the industrial partner,* • *And enhanced international cooperation and knowledge transfer between academia and industry, supporting innovation capacity in the railway sector.*

6. References

(1) **König, Jens**; Persson, Rickard (KTH Stockholm, Sweden); Lochman, Libor (Wabtec, Italy), Guicheney, Pauline (SNCF, France): Regional Rail Rolling Stock Requirements and Specifications. In: C. McNally et al. (Eds.): Transport Transitions: Advancing Sustainable and Inclusive Mobility - Proceedings of the 10th TRAconference, 2024, Dublin, Ireland - Volume 2: Sustainable Transport Development. Springer Nature Switzerland, Cham. S. 522–528. https://doi.org/10.1007/978-3-031-85578-8, 2025.

(2) Schmauder, Nicolai; Malzacher, Gregor, Fritsche, Michael; Burkat, Marcel; **König, Jens**; Boese, Ben (Alstom): High potential: lightweight optimised structural design of car bodies for railway vehicles with alternative drive systems. Discover Mechanical Engineering (3). Springer Nature Link. doi: 10.1007/s44245-024-00040-z. ISSN 2731-6564, 2024.

(3) Schmauder, Nicolai, Malzacher, Gregor, Fritsche, Michael, Burkat (Hörman Vehicle Engineering), Marcel, Boese, Ben, **König, Jens** (2023) Optimised structural design of H2 multiple units. Faculdade de engenharia - Universidade do porto. 1. International Conference on Vehicle Body Engineering, 2023-11-02 - 2023-11-03, Porto, Portugal. ISBN 978-989-917-718-5.

(4) Schmauder, Nicolai, Malzacher, Gregor (German Aerospace Center), **König, Jens**, Szengel, Konstantin, Dazer, Martin (University of Stuttgart) (2023) Development of a segmentation method as an interweaving of topology optimization and multi-body simulation. NAFEMS World Congress 2023, 2023-05-15 - 2023-05-18, Tampa, Florida, USA.

(max 10 pages)

(5) Malzacher, Gregor; Takagaki, Masakazu (RTRI, Japan); Gomes Alvers Christian; König, Jens: Comparison of design regulations for rail vehicle car body structures in Japan and Germany. (Vergleich der Auslegungsregularien für Wagenkastenstrukturen in Japan und Deutschland). In: ZEVrail - Zeitschrift für das gesamte System Bahn, 146. Georg Siemens Verlag, 2022, S 132-139

(6) **König, Jens**; Rehermann, Michael; de la Guerra Ochoa, Eduardo (Talgo, Spain): Geometric Analyses of Carbodies as the Basis for a Significant Weight Reduction in the Roll2Rail Project. In: International Railway Symposium Aachen: IRSA, 28.-30. Nov. 2017, Aachen.

(7) Malzacher, Gregor, **König, Jens**, Winter, Joachim, Vogler, Andreas (2017) AeroLiner3000 – Double-decker high speed train for GB. The Stephenson Conference, 25.-27. April. 2017, London, England.

(8) Zimmermann, Michael; **König, Jens**; Winter, Joachim; Friedrich, Horst E.: Crash concept of the Next Generation Train (NGT) intermediate cars: Fulfilment of passive safety standard EN 15227 using a full-scale demonstrator. In: Proceedings of the 11th International Symposium Passive Safety of Rail Vehicles, 62, 2017, S. 101-115.

(9) Zimmermann, Michael und **König, Jens** und Winter, Joachim und Friedrich, Horst E.: Crash Concept of the Next Generation Train Intermediate Waggon Zones. In: Civil Comp Press. Railways 2016, 06.- 08. April 2016, Cagliari, Italien. ISBN 978-1-905088-65-2 ISSN 1759-3433, 2016

(10) **König, Jens:** Novel Lightweight Concepts and Mechanical Design Methods for High Speed Rolling Stocks under Special Consideration of Car Body Lightweight Design. Doctoral Thesis University of Stuttgart. DLR-Forschungsbericht. ISRN DLR-FB--2016-33, 278 S. German Aerospace Centre. Bibliotheks- und Informationswesen, 2016.

(11) **König, Jens**, Friedrich, Horst E., Winter, Joachim, Schön, Martin (Alstom): Novel Lightweight Construction Concepts and Methods for Car Bodies. The International Journal of Railway Technology. Saxe-Coburg Publications. Railways 2014, 8.-11. April 2014, Ajaccio, France. ISSN 2049-5358, 2014

(12) Zimmermann, Michael; Beeh, Elmar; Ehrenberger, Simone; Friedrich, Horst E.; **König, Jens**; Kopp, Gundolf; Zhou, Ping: Innovative Lightweight Solutions for Road and Railway Vehicles and Life Cycle Assessment (LCA) of Magnesium. The 2nd German-Korean Workshop - on the production and industrial applications of semi-finished Mg products, S. 77-96. -. 2nd German Korean Workshop, 27.10.2014-30.10.2014, Incheon, Korea, 2014.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization